



Scientific Methodology — Study Summary

Pos-Graduação: Arquitetura e Infraestrutura de T.I. Condensed notes in English (B1 level) | March 2026

1. What is Scientific Methodology?

Scientific methodology is a **set of rules and procedures** that guides research — from finding a problem to presenting results.

Core principles:

- **Objectivity** — the researcher must be impartial
- **Verifiability** — others can check and repeat the results
- **Replicability** — the experiment can be done again under the same conditions
- **Systematicity** — research follows clear, organized steps

These principles separate scientific knowledge from common sense or personal opinion.

2. Epistemology and Scientific Methodology

At the graduate level, it is not enough to know **how** to do research. You must also understand **why** certain methods are better for certain questions.

The Three Main Paradigms

Paradigm	Reality	Focus	Methods
Positivist	Objective, external	Laws and predictions	Quantitative, experiments
Interpretivist	Socially constructed	Meanings and experiences	Qualitative, interviews
Critical	Shaped by power	Social transformation	Participatory research

Key concept: Your paradigm determines your *ontology* (what reality is) and your *epistemology* (how we can know it).

3. Types of Research Goals

- **Exploratory** — Understanding a new or little-studied topic
- **Descriptive** — Describing characteristics of a phenomenon

- **Explanatory** — Finding causes and relationships
- **Predictive** — Forecasting future behavior

Master's vs. Doctorate

Level	Main Goal
Master's (Mestrado)	Show ability to do rigorous, independent research
Doctorate (Doutorado)	Make an original and significant contribution to knowledge

4. Scientific Rigor Criteria

Criterion	Quantitative Research	Qualitative Research
Validity	Control variables strictly	Use triangulation
Reliability	Results can be replicated	Detailed documentation (audit trail)
Objectivity	Standardized procedures	Trace interpretations to raw data
Originality	New data or theory	New context or perspective

5. Types of Scientific Production

1. **Scientific Article** — Short format, published in peer-reviewed journals
2. **Master's Dissertation** — 80–150 pages, shows research competence
3. **Doctoral Thesis** — 150–300 pages, original contribution to knowledge
4. **Research Project** — Planning document, required before starting research
5. **Technical Report** — For agencies or companies, practical focus
6. **Book Chapter** — Part of a larger organized collection

6. Structure of an Academic Paper

Pre-textual Elements

Cover → Title page → Approval page → Abstract (PT + EN) → Lists → Table of contents

Textual Elements

Introduction → Theoretical Framework → Methodology → Results → Discussion → Conclusion

Post-textual Elements

References → Appendices (created by the author) → Annexes (external documents)

7. How to Write a Good Introduction

Your introduction should answer these questions:

1. What is the **broad context** of your topic?
2. What is the **specific problem** you are studying?
3. What are your **objectives** (general and specific)?
4. Why is this research **relevant**?
5. What are the **limits** of your study?

Tip: The introduction is usually 10–15% of the total work.

8. Theoretical Framework

The theoretical framework is not just a list of authors. It is the **conceptual lens** you use to analyze your topic.

Good theoretical framework:

- Selects theories that **relate directly** to your problem
- **Explains why** those theories and not others
- Shows **connections** between different concepts
- Discusses **strengths and limitations** of each theory

Usually corresponds to 25–30% of the total work.

9. Research Methodology Section

You must clearly explain:

1. **Paradigm** — positivist, interpretivist, or critical
2. **Approach** — qualitative, quantitative, or mixed
3. **Strategy** — case study, survey, experiment, ethnography, etc.
4. **Participants/Sample** — who and how many, inclusion/exclusion criteria

5. **Data Collection Instruments** — questionnaires, interviews, observations
 6. **Analysis Techniques** — statistical tools (SPSS, R) or qualitative tools (NVivo, Atlas.ti)
 7. **Ethical Aspects** — informed consent, anonymity, ethics committee approval
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10. Data Analysis

Qualitative Analysis

- Coding (open, axial, selective)
- Thematic Analysis
- Content Analysis
- Discourse Analysis
- **Tools:** NVivo, Atlas.ti, MAXQDA

Quantitative Analysis

- Descriptive Statistics (mean, standard deviation)
 - Inferential Statistics (hypothesis tests, regression)
 - **Tools:** SPSS, R, Stata, Python (Pandas, SciPy)
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11. Discussion Section

A strong discussion does the following:

1. **Recalls the main results** (briefly)
 2. **Interprets results** using the theoretical framework
 3. **Compares with previous studies** — convergences and divergences
 4. **Explains unexpected results** honestly
 5. **Acknowledges limitations** of the study
 6. **States contributions** — theoretical, methodological, practical
 7. **Suggests future research** agenda
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12. Conclusion

The conclusion should:

- Restate the **research problem**

- Show that **objectives were achieved**
- Summarize the **main findings**
- Clearly state **contributions** (theoretical, methodological, practical)
- List **limitations** honestly
- Propose a **future research agenda**

Do NOT introduce new information in the conclusion.

13. ABNT Standards (Brazilian Academic Norms)

Key norms for graduate work:

Norm	Purpose
NBR 14724:2011	General structure and formatting of academic work
NBR 6023:2018	How to write references/bibliography
NBR 10520:2023	How to write citations inside the text
NBR 6028:2021	How to write abstracts
NBR 6027:2012	Table of contents structure

Basic Formatting (NBR 14724)

- Paper: **A4**, white
 - Margins: **3cm** top/left, **2cm** bottom/right
 - Font: **Times New Roman or Arial**, size **12**
 - Line spacing: **1.5** for body text, **single** for quotations and references
 - Paragraph indent: **1.25 cm**
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14. Citations (NBR 10520:2023)

Short Direct Quote (≤ 3 lines)

Use **quotation marks** inside the paragraph. Example: According to Silva (2023, p. 45), "qualitative research allows..."

Long Direct Quote (> 3 lines)

Separate block, **4cm indent**, font size **10**, single spacing, **no quotation marks**.

Indirect Quote (Paraphrase)

Rewrite the idea in your own words. No quotation marks needed.
Example: Silva (2023) argues that qualitative research provides deeper insights...

Author-Date System

- Up to 3 authors: list all — (SILVA; SANTOS; LIMA, 2022)
- 4 or more: first author only — (SILVA et al., 2022)

15. References (NBR 6023:2018)

Book

LAST NAME, First Name. Title in bold. Edition. City: Publisher, Year.

Journal Article

LAST NAME, First Name. Article title. Journal Title in bold, City, v. X, n. X, p. XX-XX, month. year. DOI: xxxxx.

Thesis/Dissertation

LAST NAME, First Name. Title in bold. Year. Number of pages. Thesis/Dissertation (Master's/PhD in Area) — Institution, City, Year.

Online Document

ORGANIZATION. Title. Available at: URL. Access: day month year.

Important: When a work has 4 or more authors, use only the first + **et al.**

16. Useful Research Tools

Databases

Tool	Focus
Web of Science	Multidisciplinary, high impact
Scopus	Broad coverage, metrics
PubMed	Health and biomedicine

Tool	Focus
Google Scholar	Exploratory searches

Reference Managers

Tool	Cost	Best For
Zotero	Free	Flexibility, open source
Mendeley	Free (2GB)	Social/collaborative features
EndNote	Paid	Large projects, advanced control

Data Analysis Tools

- **Qualitative:** NVivo, Atlas.ti, MAXQDA
- **Quantitative:** SPSS, R, Python, Stata, JASP (free)

Systematic Review Tools

- Rayyan, Parsifal, StArt

17. Research Ethics

You must always follow these principles:

- **Honesty** — never invent, falsify, or hide data
- **Proper attribution** — always cite your sources correctly
- **Respect for participants** — get informed consent, protect privacy
- **Transparency** — document all procedures clearly
- **Declare conflicts of interest** — be open about funding or personal interests

What is Scientific Misconduct?

- **Plagiarism** — using someone's work without credit
- **Fabrication** — inventing data
- **Falsification** — changing or omitting data
- **Inappropriate authorship** — including people who didn't contribute

In Brazil, research involving humans must be approved by a **CEP (Ethics Committee)**, following resolutions **CNS 466/2012** and **CNS 510/2016**.

18. Productivity Tips for Graduate Students

- Use **Trello, Notion, or Asana** to manage your research project
- Write **daily**, even if only a little (Pomodoro technique works well)
- Use a **reference manager** from day one
- Make **regular backups** using the 3-2-1 rule:
 - 3 copies of your data
 - 2 different media
 - 1 copy off-site or in the cloud
- Attend **academic events** and build your network
- Take care of your **mental health** — postgraduate work is a marathon, not a sprint

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